measurement testing

FIRST:

1. Place a cup on the scale and **Tare (Zero)** it.
2. Run the motor for 10 seconds
3. **Write weight:**

Salt: .8mm

Output #1: 18 g

Output #2: 22 g

Output #3: 20 g

Output #4: 20 g

Output #5: 20 g

Output #6: 26 g

Output #7: 22g

Output #8: 22g

Output #9: 22g

Output #10: 22g

output #11: 24g

Output #12: 22g

Salt: .9mm

Output #1: 20 g

Output #2: 14 g

Output #3: 22 g

Output #4: 22 g

Output #5: 22 g

Output #6: 22 g

Output #7: 28 g

Output #8: 20 g

Output #9: 24 g

Output # 10: 26 g

Baking soda:

Output #1: 26 g

Output #2: 26 g

Output #3: 26 g

Output #4: 26 g

Output #5: 26 g

Output #6: 26 g

Output #7: 28 g (tiny fall)

Output #8: 28 g (fall)

Output #9: 22 g

Output #10: 28 g

Output #11: 26 g

Cinnamon:

Output #1: 8 g

Output #2: 4 g (clogged)

Output #3: 6 g

Output #4: 6 g

Output #5: 6 g

Output #6: 6 g

Output #7: 6 g

Precise measurements:

1. Run continuously until the powder starts falling out. **time that!**
2. Run the motor for exactly 10 Seconds.

- Weigh the total pile.
- Total Weight / 10 = Grams per Second (flow rate)

- Target Weight (g) / Grams per Second = Seconds to Run

Salt:
Total weight: 22 g
Flow rate: 2.2 grams/second

Target (1/4 tsp): ~1.5 grams
Seconds to run: 1.5 / 2.2 = ~ .69 seconds

Target (1/2 tsp): ~2.0 grams
Seconds to run: 2.0 / 2.2 = ~1.38 secs

Target (1 tsp): ~6.0grams
Seconds to run: 6.0 / 2.2 = ~ 2.76 secs

Target (1/2 tbsp): ~7.2 grams
Seconds to run: 7.2 /

Target (1 tbsp): ~14.4 grams
Seconds to run: 14.4 /

Baking soda:
Total weight: 26 grams
Flow Rate: 2.6 grams / second

Target (1/4 tsp): ~1.2 grams
Seconds to run: 1.2 / 2.6 = ~ 0.46 seconds

Target (1/2 tsp): ~2.4 grams
Seconds to run: 2.4 / 2.6 = ~.92 secs

Target (1 tsp): ~4.8grams
Seconds to run: 4.8 / 2.6 = ~1.85 secs

Target (1/2 tbsp): ~7.2 grams
Seconds to run: 7.2 / 2.6 = ~2.77

Target (1 tbsp): ~14.4 grams
Seconds to run: 14.4 / 2.6 = ~5.54 secs

Cinnamon:

Total weight: 6 g

Flow rate: .6 g/s

Target (1/2 tsp): ~4 grams
Seconds to run: 4 / 0.6 = 6.67 seconds

Hitting 1/4 tsp

Formula: Target Weight / grams per second = seconds to run

Salt:

Target weight: 1.5 g (1/4 tsp)

Seconds to run: ~1000 ms

1200ms

Output #1: 1/8 tsp

Output #2: 1/8 tsp

1100ms

Output #3: 1/4tsp

Output #4: overflow

Output #5: overflow

1000ms

Output #6: 1/4 (tiny under)

Output #7: 1/4 (tiny under)

Output #8: 1/8

Output #9: overflow

Output #10: 1/4 tsp

Baking powder:

Target weight: 1.2 (1/4 tsp)

Seconds to run: 1000ms

660 ms

Output #1: 1/8 tsp

Output #2: 1/8 tsp

1000 ms

Output #3: 1/4 tsp

Output #4: 1/8 tsp

Output #5: overflow

Output #6: 1/4 tsp

Output #7: 1/8 tsp

Output #8: 1/8 tsp

Output #9: overflow

tap test:

Output #1: 1/8 tsp

Baking soda:

Total weight: 1.2 g (1/4 tsp)

Seconds to run: ~seconds

460ms + 200ms

660 ms

Output #1:

Output #2:

Output #3:

Output #4:

Output #5:

Output #6:

Output #7:

Output #8:

Output #9:

Cinnamon:

Total weight: .65 g

Seconds to run:

Output #1:

Output #2:

Output #3:

Output #4:

Output #5:

Output #6:

Output #7:

Output #8:

Output #9:

Output #11: 1/4 tsp

Output #12: 1/4 tsp

Output #13: 1/4 tsp

Output #14: 1/4 tsp

Target weight: 1.5 g (1/2 tsp)

Seconds to run: ~1800 ms (above*2-.2)

Output #1: 1/2 tsp

Output #2: 1/2 tsp

Output #3: 1/2 tsp (tiny under)

Output #4: 1/2 tsp

Output #5: 1/2 tsp (little over)

Output #6: 1/2 tsp

Output #7: 1/2 tsp

Target weight: 6 grams (1 tsp):

Output #2: 1/8 tsp

Output #3: overflow

Output #4: 1/8 tsp

Output #5: overflow

Output #6: 1/8 tsp

Output #7: overflow

with agitator:

Output #1: overflow, but 1/4

Output #2: 1/8

Output #3: 1/8

Output #4: overflow, but 1/4

Output #5: 1/4

Output #6: 1/4 (tiny over)

Output #7: 1/4

Output #8: 1/4

Output #9: 1/4 (tiny over)

Output #10: 1/4

Output #11: 1/4 (tiny under)

Seconds to run: ~3400 ms (above*2-.2)

Output #1: 1 tsp

Output #2: 1 tsp

Output #3: 1 tsp

Output #4: 1 tsp (tiny over)

Output #5: 1 tsp

Output #6: 1 tsp (tiny over)

Output #7: 1 tsp

---

time = (800ms)x + (200ms)

x = amount of 1/4 tsp in the amount

Target weight: 14.4 grams (1 tbsp):

Seconds to run: ~9800 ms

Output #1: 1 tbsp

Output #2: 1 tbsp

Output #3: 1 tbsp

Target weight: 2.4 (1/2 tsp)

Seconds to run: 1800ms

Output #1: 1/2 tsp (tiny over)

Output #2: 1/2 tsp (tiny over)

Output #3: 1/2 (tiny over)

1700ms

Output #4: 1/2 tsp (teeny over)

Output #5: 1/2 tsp

Output #6: 1/2 tsp

Output #7: 1/2 tsp (teeny over)

Output #8: 1/2 tsp

Output #9: 1/2 tsp (teeny over)

Output #10: 1/2 tsp

Target weight: 4.8 (1 tsp)

Seconds to run: 3100ms

Output #1: 1 tsp

Output #4: 1 tbsp Output #5: 1 tbsp	Output #2: 1 tsp Output #3: 1 tsp Output #4: 1 tsp (tiny over) Output #5: 1 tsp Output #6: 1 tsp (tiny over) Output #7:1 tsp (tiny over) Output #8: 1 tsp (tiny under) Output #9: 1 tsp Output #10: 1 tsp (tiny under) --- Target weight: 14.4 (1 tbsp) Seconds to run: 8700 ms Output #1: 1 tbsp (tiny under) Output #2: 1 tbsp (tiny under Output #3: 1 tbsp (tiny under) 8900ms Output #4: 1 tbsp Output #5: 1 tbsp

Analyze:
If numbers are close? → done.
If wildly different → it is pulsing and need something to restrict flow or 12 flights

SALT - .8mm Tolerance: **time = (800ms)x + (200ms) → x = amount of 1/4 tsp in the amount**

time	measurement
1000 ms	1/4 tsp
1800 ms	1/2 tsp
3400 ms	1 tsp
9800	1 tbsp

BAKING SODA/POWDER - .6mm Tolerance w/ agitator

time	measurement
1000 ms	1/4 tsp
1800ms	1/2 tsp
3100ms	1 tsp
8700 ms	1 tbsp

archive:

Baking soda:

Output #1: 6 g

Output #2: 8 g

Output #3: 6 g

Output #4: 10g

Output #5:8 g

Output #6: 10 g

Output #7: 6 g

Output #8: 6 g

Output #9: 8 g